ASML Holding (ASML.AS): Pricing step-up and favourable mix drive margin outlook, while demand visibility underpins capacity roadmap

EUROPE TECHNOLOGY: HARDWARE
Analysing the role of European AI enablers and updating our view on long-term upside potential
Explore >



Alexander Duval
+44(20)7552-2995 |
alexander.duval@gs.com
Goldman Sachs International

Anant Jakhar
+1(332)245-7829 |
anant.x.jakhar@gs.com
Goldman Sachs India SPL

Ayo Odunaiya
+44(20)7051-5995 |
ayo.x.odunaiya@gs.com
Goldman Sachs International

ASML's 3Q26 revenue came in line with Visible Alpha Consensus Data and EBIT came in 5% above consensus driven by gross margins of c.57% (vs VA cons at 56%) due to stronger contributions from installed base revenues and upgrade business reflecting customers' willingness to be more productive. The company again upgraded its guidance for 2026 and now expects revenue of between €44bn and €45bn (vs. €43bn to €45bn prior) with gross margin of around 56% (vs. 54%-56% prior). As such, we note that the guide for 2026 implies revenue 1% ahead of consensus at the mid-point going into the print. Key takeaways include **1)** Long-term customer commitments reinforce demand visibility across Logic and Memory, **2)** Volume leverage and product mix drive margin progression into 2027, with upgrades a potential swing factor, **3)** Manufacturing efficiency and supply chain readiness provide scope for higher 2028 capacity, and **4)** Underlying China demand remains firm, driven by domestic Logic and Memory investment. Reiterate Buy.

- **Long-term customer commitments reinforce demand visibility across Logic and Memory:** Management reiterated that the pace of AI infrastructure investment remains strong, with supply constraints across both advanced Logic and Memory becoming more pronounced. As such, leading-edge manufacturers are providing greater demand visibility, underpinned by an increasing number of long-term commitments from their own customers, with management expecting demand to outpace supply through at least 2028. Within this backdrop, Memory is increasingly becoming an important contributor to the company's growth, fuelled primarily by DRAM investment, with FY26 Memory net system sales expected to increase c.75% yoy versus c.10% for Logic. Management also pointed to a broadening customer base for Low NA EUV, particularly in DRAM, providing further validation of the technology and adding to potential capacity requirements over 2028-30. That said, Logic demand remains robust, with advanced Logic system sales expected to grow by more than 25% in FY26. Moreover, High NA adoption roadmaps outlined by leading foundries remain encouraging, with Samsung and TSMC targeting introduction into high-volume manufacturing in 2027 and 2030, respectively.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

- **Volume leverage and product mix drive margin progression into 2027, with upgrades a potential swing factor:** The company outlined several levers of gross margin progression into 2027, namely higher system volumes, as ASML targets a >30% increase in EUV and immersion manufacturing capacity. This should provide greater fixed-cost leverage, although the extent of the benefit will depend on the company's ability to manage associated ramp costs. In parallel, product mix is expected to improve as shipments shift towards higher-productivity EUV systems, with fewer NXE:3800D models and a greater contribution from newer tools. Importantly, management confirmed that ongoing customer discussions around value-based pricing have resulted in price increases for its most critical, high-volume manufacturing products, with an initial benefit expected in 2027 and the full impact materialising in 2028. This represents an underlying pricing step-up beyond the benefits of higher system productivity and favourable mix, while upgrade pricing is also expected to evolve alongside system ASPs. Finally, management note the installed-base upgrade business as a potential swing factor for margins, seeing a sizeable remaining addressable market alongside plans to introduce new upgrade products that could further expand the opportunity. While the contribution will depend on customer demand, we see scope for incremental margin upside from upgrades, in addition to the benefits from volume, mix and pricing.
- **Manufacturing efficiency and supply chain readiness provide scope for higher 2028 capacity:** ASML reiterated confidence in its ability to increase manufacturing capacity by >30% in 2027, while evaluating scope for further >30% expansion in 2028. There is scope for capacity increases to be facilitated by optimising existing EUV and DUV manufacturing facilities, as well as reducing production cycle times as processes mature. While management acknowledged the challenges associated with scaling production, it remains confident in supply chain readiness for 2027 and believes there is sufficient time to prepare for higher output in 2028. In addition to manufacturing improvements, ASML continues to increase system productivity to deliver greater wafer capacity per tool. The next-generation NXE:3800F, expected to begin shipping early-2027, targets throughput of 260 wafers per hour versus 230 for the NXE:3800E, while new immersion systems are expected to reach 330 wafers per hour from 2027. These productivity gains may potentially allow customers to increase wafer output without a proportional increase in installed tool count, while also creating additional value-based pricing opportunities for ASML in our view.
- **Underlying China demand remains firm, driven by domestic Logic and Memory investment:** Management reiterated expectations for China to account for c.20% of FY26 net sales, noting that underlying demand remains robust across both mainstream Logic and Memory, driven by continued domestic fab construction and capacity additions. In this context, ASML expects demand from Chinese customers to remain healthy into 2027 and beyond, supported by these underlying investment requirements. On competition, management pointed to ASML's established position in the region, particularly in immersion lithography, where it maintains a very high market share. While domestic equipment development efforts continue, management sees limited evidence of meaningful competitive progress to date, though market share in the region over time may remain contingent on the extent to

which Chinese players are permitted to compete without further restrictions.

Changes to estimates

We raise our FY26-30 revenue estimates by +2% and +12% to reflect our positive view on continued strong demand for ASML's Lithography solutions (especially Litho services), in tandem with tailwinds from its planned value-based increases in ASPs. As such, we see scope for upside to our current revenue estimates from further increase in EUV production from ASML's new manufacturing facility, as well as potential for higher Immersion ASPs underpinned by value based pricing dynamics. Further, our FY26-30 gross margin estimates change between 0-1pp reflecting our latest product mix assumptions. As such, our FY26-30 EBIT estimates are c.4-12% higher, reflecting our latest topline, gross margin and opex estimates. Finally, our EPS forecasts increase by 4-12% across FY26-30 (factoring in our updated EBIT and share count assumptions).

Valuation and key risks

We are Buy-rated on ASML (on CL) with a 12-month price target of €2,315 (vs €2,200 prior) based on a 32x 2HCY27+1HCY28E P/E multiple (unchanged) applied to our higher EPS estimates. Key risks to our view and price target include EUV delays, capex cyclicality and unfavourable market share shifts.

ASML.AS	12m Price Target: **€2,315.00**	Price: **€1,528.20**	Upside: **51.5%**

Buy CL

Market cap: €586.6bn / $652.4bn
Enterprise value: €563.3bn / $626.4bn
3m ADTV: €906.9mn / $1.0bn
Netherlands
European Semiconductors, Hardware and Gaming Tech
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (€ mn) New	**32,667.3**	**44,940.1**	**60,061.0**	**79,540.8**
Revenue (€ mn) Old	32,667.3	44,013.9	58,021.9	74,328.0
EBIT (€ mn)	11,301.5	18,847.6	27,293.4	38,507.4
EPS (€) New	**24.16**	**42.20**	**59.49**	**84.48**
EPS (€) Old	24.16	38.73	57.15	78.52
EV/sales (X)	8.3	12.5	9.0	6.4
EV/EBITDA (X)	22.0	28.3	19.1	12.8
P/E (X)	30.0	36.2	25.7	18.1
Org. sales grth (%)	–	–	–	–

	6/26	9/26	12/26E	--
EPS (€)	7.47	12.58	15.27	–

Source: Company data, Goldman Sachs Research estimates, FactSet. Price as of 14 Oct 2026 close.

Disclosure Appendix

Reg AC

We, Alexander Duval, Anant Jakhar and Ayo Odunaiya, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Alexander Duval Goldman Sachs International, Anant Jakhar Goldman Sachs India SPL, Ayo Odunaiya Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for ASML Holding is/are relative to the other companies in its/their coverage universe: ASM International, ASML Holding, BE Semiconductor Industries, CD Projekt, Ericsson, Infineon, Logitech, Nebius Group, Nokia, STMicroelectronics, Stillfront, Technoprobe S.p.A.

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs has received compensation for investment banking services in the past 12 months: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

Goldman Sachs had a non-investment banking securities-related services client relationship during the past 12 months with: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

Goldman Sachs is a registered market maker in the securities or derivatives thereof: ASML Holding (€1,528.20) and ASML Holding NV (ADR) ($1,650.00)

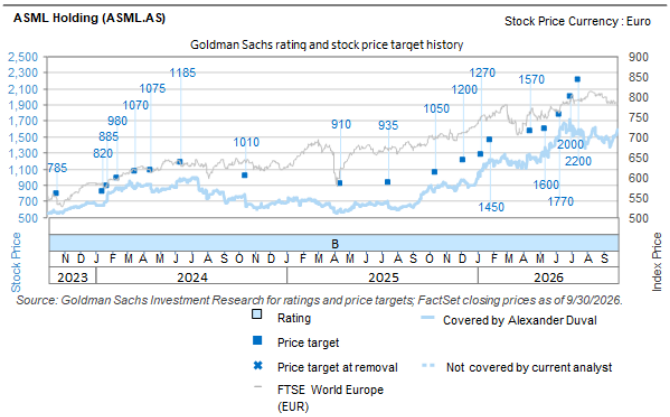
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

ASML Holding (ASML.AS)

Date of report	Target price (€)	Closing price (€)
16-Jul-26	2,200.00	1,598.40
01-Jul-26	2,000.00	1,641.60
10-Jun-26	1,770.00	1,507.20
13-May-26	1,600.00	1,327.00
16-Apr-26	1,570.00	1,222.60
29-Jan-26	1,450.00	1,192.00
12-Jan-26	1,270.00	1,086.40
08-Dec-25	1,200.00	963.20
16-Oct-25	1,050.00	877.40
17-Jul-25	935.00	650.20
16-Apr-25	910.00	574.00
16-Oct-24	1,010.00	633.90
14-Jun-24	1,185.00	953.00
17-Apr-24	1,075.00	852.40
18-Mar-24	1,070.00	870.80
12-Feb-24	980.00	877.60
24-Jan-24	885.00	775.80
15-Jan-24	820.00	648.20
18-Oct-23	785.00	553.20

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or

policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.